

AUTOMOTIVE INDUSTRY STANDARD

Uniform Provisions Concerning the Approval of Vehicles with regards to Event Data Recorder (EDR)

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT and HIGHWAYS
GOVERNMENT OF INDIA

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INTRODUCTION

The Government of India felt the need for a permanent agency to expedite the publication of standards and development of test facilities in parallel when the work on the preparation of the standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standards Committee (AISC) vide order No. RT-11028/11/97-MVL dated September 15, 1997. The standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CMVR-TSC). After approval, the Automotive Research Association of India, (ARAI), Pune, being the Secretariat of the AIS Committee, will publish this standard. For better dissemination of this information ARAI may publish this document on their Website.

The intention of this standard is to establish uniform provisions concerning the approval of motor vehicle of the categories M2, M3, N2 and N3 with regard to their Event Data Recorders (EDR's)

The provisions concern the minimum collection, storage and crash survivability of the motor vehicle crash event data. It does not include specifications for data retrieval tools and methods which are subject to national or regional level requirements.

The purpose of these provisions is to ensure that EDRs record, in a readily usable manner, data valuable for effective crash investigations and for analysis of safety equipment performance while limiting, to the greatest extent possible, the recording of data unrelated to the crash. Such crash data will help provide a better understanding of the circumstances in which crashes and injuries occur and will facilitate the development of safer vehicle designs. In this context, crashes should be understood as involving property damage and / or personal harm, including that of vulnerable road users involved.

It is understood that, in the current state of technology, the aforementioned objective can be reached only by recording the data in a specified time period based on defined triggers and trigger thresholds. These triggers may, but do not always immediately precede or follow or coincide with the crash.

While formation of this standard reference is drawn from UN R 169. Composition of the Panel and Automotive Industry Standards Committee (AISC) responsible for preparation and approval of this standard are given in Annexure 5 & Annexure 6 respectively

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**Uniform provisions concerning the approval of vehicles
with regards to Event Data Recorder (EDR)**

1.0 SCOPE

- 1.1 This standard applies to the approval of vehicles of categories M2, M3, N2 and N3 with regard to their Event Data Recorders (EDRs).
- 1.2 This standard is without prejudice to requirements of national or regional laws related to privacy, data protection and personal data processing.
- 1.3 The following data elements are excluded from the scope: Vehicle Identification Number (VIN), associated vehicle details, location / positioning data, information on the driver, and date and time of an event.
- 1.4 If there is no system or sensor designed to provide the trigger indicated in the 5.3.1.3 table of safety systems or the data elements to be recorded and stored under clause 5, in the format (range, resolution, and sample rate) indicated in Annexure 4, “Data Elements and Format” or it is not operational at the time of reaching a specific trigger condition as indicated in 5.3.1 or at the time of recording, this document requires neither recording of such data nor fitting or making such systems or sensors operational. However, if the vehicle is fitted with an original equipment manufacturer sensor or system designed to provide the trigger indicated in 5.3.1.3 or the data element in the format specified in Annexure 4, “Data Elements and Format”, then it is mandatory to report the data element in the specified format when the sensor or system is operational. In the case the reason for not being operational at the time of recording is a failure of this system or sensor, this failure state shall be recorded by the EDR as defined in Annexure 4, “Data Elements and Format”.

2.0 DEFINITIONS

For the purpose of this standard:

- 2.1 “Accelerator pedal position” means the actuation of the device that indicates the driver’s demand for acceleration to the propulsion system given in percentage of measured range of the device. This can also cover one pedal drive that may be demanding negative torque or even service braking in lower ranges.
- 2.2 “Antilock braking system” means a system which detects wheel slip and automatically modulates the pressure producing the braking forces at the wheel(s) to limit the degree of wheel slip.
- 2.3 “Antilock brake system status – tractor” indicates the status of the antilock brake system on the vehicle / tractor.
- 2.4 “Antilock brake system status – trailer” indicates the status of the antilock brake system on trailer(s).
- 2.5 “Advanced Emergency Braking System” means a system which can automatically detect a potential forward collision and activate the vehicle braking system to decelerate the vehicle with the purpose of avoiding or mitigating a collision. The system may also be referred to as “Automatic Emergency Braking System”.

- 2.6 “Brake Status Parking” indicates the status of the switch that is installed to detect whether or not the parking brake has been applied.
- 2.7 “Brake Status Service” indicates the status of the switch that is installed in the brake system to detect whether the service brake has been applied.
- 2.8 “Vehicle Stability Function” means vehicle stability control as defined by IS 11852: 2019 or AIS-150, as amended from time to time. The system may also be referred to as “Electronic Stability Control”.
- 2.9 “Propulsion System Activation Hours” means the accumulated time that the propulsion system is activated including idle state.
- 2.10 “Propulsion System Torque” means the percentage of peak or reference torque.
- 2.11 “Propulsion System Power” means the instantaneous power provided by the propulsion system.
- 2.12 “Propulsion System Drive Speed” means the rotational speed of the output shaft of the propulsion system.
- 2.13 “Event” means a crash or other physical occurrence that causes the trigger threshold to be met or exceeded.
- 2.14 “Event Data Recorder (EDR)” means a device or function in a vehicle, the purpose of which is to record the vehicle’s dynamic, time – series data during the time period just prior to, during, and after an event (e.g. vehicle speed versus time).
- 2.15 “Event Data Recording Complete” indicates whether a complete set of data that the event data recording device is designed to capture was successfully recorded by and stored in the device.
- 2.16 “End of Event Time” means the moment at which the cumulative change in velocity within a 20 ms time period becomes 0.8 km/h or less.
- 2.17 “EDR unit hardware part number” means the part number for the EDR unit.
- 2.18 “EDR unit software part number” means the part number / version number for the EDR software.
- 2.19 “Ignition Cycle Event” means the number (count) of power mode cycles as determined by the EDR ECU at the time when the event occurred since the first use of the EDR.
- 2.20 “Ignition Cycle Download” means the number (count) power mode cycles as determined by the EDR ECU at the time when the data was downloaded since the first use of the EDR.
- 2.21 “Lane Departure Warning System Status” indicates the status of the lane departure warning system.
- 2.22 Maximum Delta V, Longitudinal: means the maximum value of the cumulative change in velocity along the longitudinal axis of the vehicle within a time period of

300 ms after time zero or before the end of event time plus 30 ms, whichever is shorter.

- 2.23 “Maximum Delta V, lateral” means the maximum value of the cumulative change in velocity along the lateral axis of the vehicle within a time period of 300ms after time zero or before the end of event time plus 30ms, whichever is shorter.
- 2.24 “Maximum Delta V, resultant” means the time correlated maximum value of the cumulative change in velocity, as reported by the EDR along the vector – added longitudinal and lateral axis.
- 2.25 “Time for maximum delta V,” means the time from time zero to the point where the maximum value of the cumulative change in velocity, as recorded by the EDR, is found.
- 2.26 “Retarder torque mode” indicates which function is currently generating, limiting, or controlling the retarder torque.
- 2.27 “Roll angle” means the angle around which the vehicle rotates about its longitudinal axis.
- 2.28 “Rollover protection control System” means the stability control of brakes for rollover protection.
- 2.29 “Roll Rate” means the change of angle per unit time at which the vehicle rotates about its longitudinal axis.
- 2.30 “Supplemental Restraint System” means a passive safety system as declared by the vehicle manufacturer, supplementing the restraint system as defined by IS 15140, with components such as airbags or seatbelt pre-tensioners.
- 2.31 “Safety Belt Status” means the feedback from the safety system that the vehicle’s safety belt is fastened or not fastened.
- 2.32 “Steering Wheel Angle” means the angle of the steering shaft connected to the driver control.
- 2.33 “System Intervention” means the activation of a system, as defined by the manufacturer.
- 2.34 “Parking Brake System” means the parking braking system as regulated by AIS-150 or IS 11852: 2019, as amended from time to time
- 2.35 “Time Zero” is the time reference for the EDR data timestamps of an event.
- 2.36 “Tyre pressure monitoring system status” means the operating status of the Tyre Pressure Monitoring System.
- 2.37 “Trigger thresholds” means the appropriate parameter has met the conditions for recording an EDR event.
- 2.38 “Trigger activated” indicates which trigger was activated to cause the recording of the event.

- 2.39 “Stability Control System Status – Fully Operational” indicates whether Stability Control is fully operational or whether its functionality is reduced by a permanent or temporary (e.g. low voltage) defect, by intended action (e.g. disabled by a switch or during special diagnostic procedures), not configured or not yet fully initialized (e.g. missing initialization or configuration message).
- 2.40 “Vehicle Master Control Switch” means the device by which the vehicle’s on-board electronics system is brought, from being switched off, as in the case where a vehicle is parked without the driver being present, to normal operation mode.
- 2.41 “Vehicle Speed” means the longitudinal speed of the vehicle that is calculated or estimated from the Vehicle Speed Sensor (VSS).
- 2.42 “Vehicle model and its variants with regards to its Event Data Recorder” means vehicles which do not differ significantly in such essential aspects as:
- (a) The manufacturer’s trade name or mark
 - (b) Vehicle features which significantly influences the performances of the EDR; Addition of new trigger(s), new data (elements), or modification in their format, shall not be considered as “significantly” influencing the performance of EDR”.
 - (c) The main characteristics and design of the EDR
- 2.43 “Yaw Control System” means the stability control of wheel brakes for yaw control.
- 2.44 “Corrective Steering Function (CSF) status” means the operating status of the Corrective Steering Function as defined in AIS-193.
- 2.45 “Emergency Steering Function (ESF) Status” means the operating status of the Emergency Steering Functions as defined in AIS-193.
- 2.46 “Automatically Commanded Steering Function (ACSF) status” means the operating status of the Automatically Commanded Steering Function as defined in AIS-193, and of the categories as defined therein.
- 2.47 “Accident Emergency Call System Status” means the status of a system that is activated either automatically via in vehicle sensors or manually, which carries, by means of public mobile wireless communications network, a set of crash related data and establishes an emergency audio channel between the occupants of the vehicle and an answering point. Faulted status would mean when the system detects it cannot make a call.

3.0 APPLICATION FOR APPROVAL

- 3.1 The application for approval of a Vehicle model and its variants with regard to its EDR shall be submitted by the vehicle manufacturer or by its authorized representative to the Test Agency.
- 3.2 It shall be accompanied by the following documentation (a model of the information document is given in Annexure 2).

- 3.2.1 A description of the Vehicle model and its variants with regard to the items specified in clause 5, below, in particular related to the location of the EDR in the vehicle, the triggering parameters, storing capacity and the resistance to high deceleration and mechanical stress of a severe impact.
- 3.2.2 The data elements and format stored in the EDR;
- 3.2.3 Instructions for retrieving the data from the EDR.
- 3.3 A vehicle representative of the Vehicle model and its variants to be approved shall be submitted to the Test Agency responsible for conducting the approval tests.

4.0 APPROVAL

- 4.1 If the Vehicle model and its variants submitted for approval pursuant to this standard meets the requirements of clause 5, below approval of that Vehicle model and its variants shall be granted.
- 4.1.1 Vehicles referred to in clause 4.1, which are not subject to national and regional standards providing for crash tests requirements referred to in clause 5.4.1 and 5.4.2 of, AIS-192, as amended time to time, shall be subject to clause 5.4 of this standard.

5.0 SPECIFICATIONS

Requirements for vehicles fitted with an EDR include data elements, data format and data capture.

- 5.1 Data Elements
 - 5.1.1 Each vehicle fitted with an EDR shall record the data elements specified as mandatory and those required under specified minimum conditions during the interval / time and at the sample rate specified in Annexure 4.
- 5.2 Data format
 - 5.2.1 Each data element recorded shall be reported as specified in Annexure 4, Table 1.
- 5.3 Data Capture

The EDR shall capture data which shall be written to a non – volatile memory when any of the triggers in clause 5.3.1 occur.

The EDR shall record the captured data in the vehicle and this data shall remain in the vehicle subject to the provisions of clause 5.3.4, at least until they are retrieved in compliance with national or regional legislation, or they are overwritten in compliance with clause 5.3.4.

The EDR non-volatile memory buffer shall accommodate the data related to at least five different events.

The data elements for every event shall be captured and recorded by the EDR as specified in clause 5.1, in accordance with the following conditions and

circumstances;

5.3.1 Conditions for triggering recording of data:

As event shall be recorded by the EDR if one of the following threshold values is met or exceeded. Triggers that occur such that an overlap of data between events would result may be excluded.

5.3.1.1 Sudden Deceleration: Vehicle speed changes at a rate higher than 3.25 m/s^2 and the change persists beyond that threshold for at least 0.7 seconds.

5.3.1.2 Last Stop: Trigger shall be activated if any of the following applies:

(a) Vehicle Speed is reported as 0 km/h for 20 seconds

(b) While vehicle speed is reported as 0 km/h, and

(i) Parking brake system is applied, or

(ii) Vehicle master control switch is deactivated

Re-activation of last stop trigger due to threshold criterion (a), shall be disabled if the vehicle speed is not reported as 24 km/h or more for a minimum of 6 seconds.

5.3.1.3 Activation of a safety system is showed in the table below:

System (if fitted)	Trigger
Supplemental Restraint System	Deployment Command of a Supplement Restraint System
Antilock Braking System	System Intervention
Advanced Emergency Braking (including pedestrian / cyclist if equipped)	Emergency Brake Intervention
Vehicle Stability Function	System Intervention

5.3.2 Conditions for locking of data

In all the cases with supplemental restraint system activation, the memory for the event shall be locked to prevent any future overwriting of the data by subsequent event.

5.3.3 Conditions for the establishment of time zero.

Time zero is established by the occurrence of any of the above triggers, except for the last stop.

Time zero for the last stop trigger is established when the vehicle reaches an indicated speed of 0 km/h.

5.3.4 Overwriting

5.3.4.1 If an EDR non-volatile memory buffer void of previous event data is not available, the recorded data shall, subject to the provisions of clause 5.3.2, be overwritten by the current event data, on a first in first out basis, or according to the relevant authorities.

5.3.4.2 Furthermore, if an EDR non-volatile memory buffer void of previous event data is not available, data originating from supplemental restraint system events shall always overwrite any other data that is not locked per clause 5.3.2.

5.3.5 Power and Communication Failure

Data recorded in the non -volatile memory is retained after a loss of power. However, data need not be recorded when the power or the communication is lost to the device hosting the EDR or systems providing data.

5.4 Survivability

5.4.1 The data elements listed in Annexure 4 shall be retrievable in the format specified even after an impact. Therefore, event data recorders shall resist internal loads which may occur during a vehicle crash and be mounted in the vehicle in a position of sufficient structural integrity to protect against physical damage due to front and side impacts that would prevent the retrieval of data. To demonstrate these capabilities, Option 1 or Option 2 applies at the choice of the manufacturer.

Option 1:

EDR's shall withstand mechanical shocks at a severity level as specified in the component test of ANNEXURE-IX-C of the AIS-038 (Rev. 2), as amended time to time. The devices shall be connected to the test fixture only by the intended mountings provided for the purpose of attaching the event data recorders to the vehicle and in an orientation representative of the vehicle installation.

EDR device(s) shall be mounted in the vehicle cab / passenger compartment or in a position of sufficient structural integrity to protect against physical damage (mechanical integrity) that would prevent the retrieval of data at least in front and side impacts of a severity level corresponding to the mechanical shock requirements above. For positions outside the vehicle cab / passenger compartment, the sufficient structural integrity shall be demonstrated to the technical service together with appropriate documentation (e.g. calculations or simulations).

Option 2:

The manufacturer demonstrates that data is retrievable even after an impact of a severity level set by AIS-098 (Annex 1), AIS-099 (Annex 1) or AIS-201 (Annex 1).

5.5 It shall not be possible to deactivate the Event Data Recorder.

6.0 MODIFICATION OF VEHICLE MODEL AND ITS VARIANTS AND EXTENSION OF APPROVAL

- 6.1 Every modification of the Vehicle model and its variants as defined in clause 2.42 of this standard shall be notified to the test agency which approved the Vehicle model and its variants. The test agency may then either:
 - 6.1.1 Consider that the modifications made do not have an adverse effect on the conditions of the granting of the approval and grant an extension of approval;
 - 6.1.2 Consider that the modifications made affect the conditions of the granting of the approval, and require further tests or additional checks before granting an extension of an approval.
- 6.2 Confirmation or refusal of approval, specifying the alterations, shall be communicated by the procedure specified in clause 4.3 above to the Test Agency of the agreement applying this standard.

ANNEXURE I
RESERVED

ANNEXURE 2						
INFORMATION DOCUMENT ON THE TYPE APPROVAL OF A VEHICLE MODEL AND ITS VARIANTS WITH REGARD TO ITS EVENT DATA RECORDER						
	A list of contents shall be included					
	Any drawings shall be supplied in an appropriate scale and in sufficient detail on size A4 paper or on a folder of A4 format.					
	Photographs, if any, shall show sufficient detail					
	General					
1.0	Trade name or mark of vehicle					
2.0	Vehicle model and its variants					
3.0	Means of identification of type, if marked on the vehicle					
4.0	Location of the marking					
5.0	Location of and method of affixing the approval mark					
6.0	Category of vehicle					
7.0	Name and address of manufacturer					
8.0	Address (es) of assembly plant (s)					
9.0	Photograph (s) and / or drawing (s) of a representative vehicle					
10.0	EDR					
10.1	Make (trade name of manufacturer)					
10.2	Type and general commercial description (s)					
10.3	Drawing(s) or photographs showing the location and method of attachment of the EDR in the vehicle					
10.4	Description of the triggering parameter					
10.5	Description of any other relevant parameter (Storing capacity, resistance to high declaration and mechanical stress of a severe impact, etc.)					
10.6	The data elements and data format stored in the EDR.					
	Data elements	Recording interval / time (relative	Data sample rate (samples	Minimum range	Accuracy	Resolution

		to trigger event)	per second)			
10.7	Instructions for retrieving the data from the EDR					

ANNEXURE 3
RESERVED

ANNEXURE 4

DATA ELEMENTS AND FORMAT

The data elements specified in Table 1 shall be recorded by the Event Data Recorder if mandated under CMVR Refer clause 1.4 of scope

Table 1 – List of Data Elements ¹

<i>Data element</i>	<i>Condition for requirement ⁽²⁾</i>	<i>Recording interval / time ⁽³⁾ (relative to time zero)</i>	<i>Data sample rate (sample per second)</i>	<i>Minimum range</i>	<i>Accuracy ⁽⁴⁾</i>	<i>Resolution</i>	<i>Data recorded for the following triggers</i>
Event Data Recording Complete	Mandatory	Following other data	N/A	Yes or No	N/A	Yes or No	All 5.3.1 triggers
Propulsion system activation hours event	Mandatory ⁵	-1.0 sec	N/A	0 to 1,193,046 hr	±0.05 hr	0.05 hr	All 5.3.1 triggers
Propulsion system activation hours download	Mandatory ⁶	At time of download	N/A	0 to 1, 193, 046 hr	± 0.05 hr	0.05hr	All 5.3.1 triggers
EDR unit hardware part number	Mandatory ⁷	N/A	N/A	N/A	N/A	N/A	N/A
EDR unit software part number	Mandatory ⁷	N/A	N/A	N/A	N/A	N/A	N/A

¹ Format requirements specified below are minimum requirements and manufacturers can exceed them.

². “Mandatory” is subject to the conditions detailed in clause 1

³. Pre-Crash data and crash data are asynchronous. The sample time accuracy requirement for pre-crash time is -0.1 to 1.0 sec (e.g. T =-1 would need to occur between -1.1 and 0 seconds)

⁴. Accuracy requirement only applies within the range of the physical sensor, if measurements captured by a sensor exceed the design range of the sensor, the reported element shall indicate when the measurement first exceeded the design range of the sensor.

⁵. Manufacturers shall record either Propulsion system activation hours event or ignition cycle, event.

⁶. Manufacturers shall record either propulsion system activation hours download or ignition cycle, download.

⁷. Shall not contain unique serial numbers or other unique identifiers. If serial number traceability is integral to part number – it does not need to be reported.

Data element	Condition for requirement⁽²⁾	Recording interval / time⁽³⁾ (relative to time zero)	Data sample rate (sample per second)	Minimum range	Accuracy⁽⁴⁾	Resolution	Data recorded for the following triggers
Ignition cycle, event	Mandatory ⁵	-1.0 sec	N/A	0 to 60,000	± 1 cycle	1 cycle	All 5.3.1 triggers
Ignition cycle, download	Mandatory ⁶	At time of download	N/A	0 to 60,000	± 1 cycle	1 cycle	All 5.3.1 triggers
Vehicle Speed	Mandatory	-20 to +10 sec	4	0 to 250 km/h	±1 km/h	1 km/h	All 3.5.1 triggers
Gear Position Status	Mandatory ⁹	-20 to +10 sec	4	N/A	N/A	Reverse	All 3.5.1 triggers
Retarder Torque Mode	Mandatory / if fitted*	-20 to +10 sec	4	N/A	N/A	Default, Operator Selection, Cruise Control, Road Speed Limit, Stability Control, Transmission Control, Engine Speed Limit, Braking System	All 5.3.1 triggers
Brake Status – Parking	Mandatory	-20 to +10 sec	4	N/A	N/A	On or Off	All 5.3.1 triggers
Brake Status Service	Mandatory	-20 to +10 sec	10	N/A	N/A	On or Off	All 5.3.1 triggers
⁸ Event indicates recording around the time of the trigger.							
⁹ Gear position is mandatory if the vehicle speed data element only records forward speeds.							

<i>Data element</i>	<i>Condition for requirement⁽²⁾</i>	<i>Recording interval / time⁽³⁾ (relative to time zero)</i>	<i>Data sample rate (sample per second)</i>	<i>Minimum range</i>	<i>Accuracy⁽⁴⁾</i>	<i>Resolution</i>	<i>Data recorded for the following triggers</i>
Propulsion system Torque ¹⁰	Mandatory	-20 to +10 sec	4	Actual value reported in Nm	<100Nm ±10 Nm and >100 Nm: 5% of torques value which was recorded	1Nm	All 5.3.1 triggers
Propulsion system power ¹⁰	Mandatory	-20 to +10 sec	4	Actual value reported in kW	±5%	1kW	All 5.3.1 triggers
Propulsion system drive speed ¹⁰	Mandatory	-20 to +10 sec	4	Actual value reported in rpm	±100 rpm	100 rpm	All 5.3.1 triggers
Accelerator Pedal Position	Mandatory	-20 to +10 sec	10	0 to 100%	± 5	1%	All 5.3.1 triggers
ABS Brake Control Status – Motor vehicle	Mandatory if fitted*	-20 to +10 sec	10	N/A	NA	Off, ABS Passive but installed, ABS Active	All 5.3.1 triggers
ABS Brake Control Status – Trailer	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Off, Faulted, On not intervening On intervening	All 5.3.1 triggers
Advanced Emergency Braking	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Off, Faulted, On not warning / intervening , On warning, On intervening	All 5.3.1 triggers
Lane Departure Warning System Status	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On not warning, On – Warning	All 5.3.1 triggers
Steering wheel angle	Mandatory if fitted*	-20 to 10 sec	10	-1776 degrees to +1776 degrees	± 0.4 rad 22.9 degrees	0.2 rad 11.5 degrees	All 5.3.1 triggers
Stability Control System Status	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Fully operational Not fully operational	All 5.3.1 triggers.

¹⁰. if motors or other drivers are available as separate items then these should be listed with relevant location, e.g. 1st left or 2nd left, 1st right or 2nd right (driven axles), 1st or 2nd, nth (for combined drivers) for speed, torque and power. For hybrid systems, engine and motor elements should be listed separately.

Data element	Condition for requirement⁽²⁾	Recording interval / time⁽³⁾ (relative to time zero)	Data sample rate (sample per second)	Minimum range	Accuracy⁽⁴⁾	Resolution	Data recorded for the following triggers
Rollover Protection Control System Status	Mandatory if fitted*	-20 to +10 sec	4	N/A	N/A	Passive but installed Active	All 5.3.1 triggers
Yaw Control system status	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Passive but installed, Active	All 5.3.1 triggers
Safety belt status (position x-y) ¹¹	Mandatory if fitted*	-1.0 sec	N/A	N/A	N/A	Fastened, not fastened	All 5.3.1 triggers
Safety belt pre-tensioner status (position x-y) ¹¹	Mandatory if fitted*	Event ⁸	N/A	N/A	N/A	Faulted, not deployed, deployed	Supplemental restraint system
Frontal Airbag system status ¹¹	Mandatory if fitted*	Event ⁸	N/A	N/A	N/A	Faulted, suppressed (passenger), deployed not deployed	Supplemental restraint system
Side Airbag system status ¹¹	Mandatory if fitted*	Event ⁸	N/A	N/A	N/A	Faulted deployed, not deployed	Supplemental restraint system
Side curtain / Tube air bag system status ¹²	Mandatory if fitted*	Event ⁸	N/A	N/A	N/A	Faulted, deployed, not deployed	Supplemental restraint system
Far-side impact centre air bag system status ¹²	Mandatory if fitted*	Event ⁸	N/A	N/A	N/A	Faulted, deployed, not deployed	Supplemental restraint system
Blind Spot Information System for the detection of bicycles	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Off, Faulted, On not warning, On warning left-side. On warning right side.	All 5.3.1 triggers
Reversing motion VRU detection system	Mandatory if fitted*	-20 to +10sec	10	N/A	N/A	Off, Faulted On not warning, On warning	All 5.3.1 triggers

¹¹. this data element shall be recorded for all seating positions equipped with systems and sensors in accordance with clause 1.4 A separate row shall be added to the report for each seating position, and the position shall be denoted as 'x=seat row number, starting with 1 at the vehicle front, y= seat number, starting with 1 at the vehicle's left side. For example, 'position 1-1 denotes the front leftmost seat and position 1-2 denotes the front second from the left.

¹². list this element n times, once for each airbag.

<i>Data element</i>	<i>Condition for requirement⁽²⁾</i>	<i>Recording interval / time⁽³⁾ (relative to time zero)</i>	<i>Data sample rate (sample per second)</i>	<i>Minimum range</i>	<i>Accuracy⁽⁴⁾</i>	<i>Resolution</i>	<i>Data recorded for the following triggers</i>
Moving Off information system for the Detection of Pedestrian and cyclists	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Off, Faulted, On not warning, On warning	All 5.3.1 triggers
Maximum delta – V, longitudinal	Mandatory if fitted*	Event ⁸	N/A	-100 km/h to +100 km/h	± 10%	1km/h	Supplemental restraint system (planar events)
Time, maximum delta- V, longitudinal	Mandatory if fitted*	Event ⁸	N/A	0-300 ms	± 3ms	2.5ms	Supplemental restraint system (planar events)
Maximum delta- V, Lateral	Mandatory if fitted*	Event ⁸	N/A	-100 km/h to +100 km/h	± 10%	1km/h	Supplemental restraint system (planar events)
Time for maximum delta-V, lateral	Mandatory if fitted*	Event ⁸	N/A	0-300 ms	±3ms	2.5 ms	Supplemental restraint system (planar events)
Maximum delta-V resultant	Mandatory if fitted*	Event ⁸	N/A	-100 km/h to +100 km/h	± 10%	1 km/h	Supplemental restraint system (planar events)
Time for maximum delta – V, resultant	Mandatory if fitted*	Event ⁸	N/A	0-300 ms	± 3ms	2.5 ms	Supplemental restraint system (planar events)
Roll angle	Mandatory if fitted*	-20 to +10 sec	4	-1080 deg to +1080 deg	± 10%	10 deg	Supplemental restraint system (rollover events)

Data element	Condition for requirement⁽²⁾	Recording interval / time⁽³⁾ (relative to time zero)	Data sample rate (sample per second)	Minimum range	Accuracy⁽⁴⁾	Resolution	Data recorded for the following triggers
Roll rate	Mandatory if fitted and used for rollover occupant protection system control algorithm	-20 to +10 sec	4	-240 to +240 deg/sec	± 10%	4 deg/sec	Supplemental restraint system (rollover events)
Longitudinal Acceleration	Mandatory if fitted*	-20 to +10 sec	4	-1.5g to +1.5g	±10%	0.1g	All 5.3.1 triggers
Lateral acceleration	Mandatory if fitted*	-20 to +10 sec	4	-1.5g to +1.5g	± 10%	0.1g	All 5.3.1 triggers
Accident emergency call system status	Mandatory if fitted*	Event ⁸	N/A	N/A	N/A	Faulted, On but emergency call not automatically triggered, On-Emergency call automatically triggered	Supplemental restraint system
Tyre pressure monitoring system warning lamp status	Mandatory if fitted*	-20 to +10 sec	4	N/A	N/A	On, Off	All 5.3.1 triggers
Yaw Rate	Mandatory if fitted*	-20 to +10 sec	4	-75 to +75 degrees / second	± 10% of the full range of the sensor	1 degree per second	Supplemental restraint system (planar events)
Corrective steering function status ¹³	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On but not intervening On actively intervening	All 5.3.1 triggers
Emergency steering function status ¹³	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On but not intervening, On actively intervening	All 5.3.1 triggers

¹³. Only applies to vehicles subject to approval in accordance with any UN Regulation annexed to the 1958 agreement with respect to their steering functions.

Data element	Condition for requirement⁽²⁾	Recording interval / time⁽³⁾ (relative to time zero)	Data sample rate (sample per second)	Minimum range	Accuracy⁽⁴⁾	Resolution	Data recorded for the following triggers
Automatically commanded steering function category A status ¹³	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On but not controlling, On controlling	All 5.3.1 triggers
Automatically commanded steering function category B status ¹³	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On but not controlling, On controlling	All 5.3.1 triggers
Automatically commanded steering function category C status ¹³	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On but not controlling, On controlling	All 5.3.1 triggers
Automatically commanded steering function category D status ¹³	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On but not controlling, On controlling	All 5.3.1 triggers
Automatically commanded steering function category E status ¹³	Mandatory if fitted*	-20 to +10 sec	10	N/A	N/A	Faulted, Off, On but not controlling, On controlling	All 5.3.1 triggers

* Note: Mandatory provisioning of this data element shall be in accordance with the notified provisions under CMV(A) R, 1989. As and when such provisions are made mandatory under CMVR, by the Statutory Authority, the data elements for such respective provisions shall be provided by the Vehicle manufacturer and shall be captured by EDR.

ANNEXURE 5
(See Introduction)
COMPOSITION OF AISC PANEL ON EVENT DATA RECORDER (EDR)

Convener	
Shri A. A. Badusha	The Automotive Research Association of India
MEMBERS	REPRESENTING
Shri Manoj Desai	The Automotive Research Association of India
Shri Kamalesh Patil	The Automotive Research Association of India
Shri Avik Chatterjee	The Automotive Research Association of India
Shri N. Ramesh	The Automotive Research Association of India
Shri Pranab Devrajan	The Automotive Research Association of India
Shri Vignesh Kumar	The Automotive Research Association of India
Shri Pratik Nayak	The Automotive Research Association of India
Shri S. Perumal	Global Automotive Research Centre
Shri Saurabh	National Automotive Test Tracks
Shri N. Muthukumar	Ashok Leyland Ltd., (SIAM)
Shri V. Faustino	Ashok Leyland Ltd., (SIAM)
Shri V. G. Kulkarni	Mahindra & Mahindra Ltd. (SIAM)
Shri Akshay A. P	Mahindra & Mahindra Ltd. (SIAM)
Shri Uday Salunkhe	Tata Motors Ltd. (SIAM)
Shri Ajay Pandey	Tata Motors Ltd. (SIAM)
Shri Rama Manikandan	DICV (SIAM)
Shri Venkatesh G	DICV (SIAM)
Shri Jebin Jowar	RNTBCI (SIAM)
Shri Chandrashekara J	Volvo Trucks (SIAM)
Shri Ganesh L P	Bosch (ACMA)
Shri Vinayak Patil	Bosch (ACMA)

Shri Praveen Chilshetty	Knorr Bremse (ACMA)
Shri Ruban Devaprasath	Brakes India Ltd (ACMA)
Shri G. Devendran	Brakes India Ltd (ACMA)

ANNEXURE 6
(See Introduction)

COMMITTEE COMPOSITION *
Automotive Industry Standards Committee

Chairperson	
Dr. Reji Mathai	Director, The Automotive Research Association of India
Members	Representing
Representative from	Ministry of Road Transport and Highways
Representative from	Ministry of Heavy Industries
Representative from	Office of the Development Commissioner, MSME, Ministry of Micro, Small and Medium Enterprises
Head-TED	Bureau of Indian Standards
Director	Central Institute of Road Transport
Director	Global Automotive Research Centre
Director	International Centre for Automotive Technology
Director	Indian Institute of Petroleum
Director	Vehicles Research and Development Establishment
Director	Indian Rubber Manufacturers Research Association
Representatives from	Society of Indian Automobile Manufacturers
Representatives from	Tractor and Mechanization Association
Representatives from	Automotive Components Manufacturers Association of India
Representative from	Indian Construction Equipment Manufacturers' Association
Member Secretary	
Shri Vikram Tandon	The Automotive Research Association of India

* At the time of approval of this Automotive Industry Standard (AIS)